

LCMU Cal Data Capture

Bench Tool User Guide

Ken Overton

KJO Liquid Rocket Motor Control System

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Standalone bench-test firmware, flashed onto LCMU's physical hardware, that logs raw readings from all three load cells to the SD card on a button press, for deriving a platform-level weight calibration curve.

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1 Purpose

This tool captures one reading from each of LCMU's three load cells to a CSV-style file on the SD card every time Button C is pressed, timestamped from the real-time clock. It is meant for bench calibration work: apply a known reference weight to the platform, press Button C to log the raw per-cell readings (and their sum), repeat at several reference weights, then fit a calibration curve from the logged data offline.

It runs on the same physical LCMU hardware as production firmware, but is a completely separate, standalone build – **no CAN**; it never communicates with EMU or GSEMU.

2 Hardware Required

- LCMU hardware: Adafruit Feather RP2040 Adalogger, SH1107 OLED, DS3231 RTC, PCA9548 I²C mux, 3×NAU7802 load-cell ADCs, onboard microSD
- All three load cells connected and seated on the platform
- Button C wired (direct GPIO on this board – no I²C expander, unlike EMU's equivalent tool)

3 How to Use

1. Power on the board. The OLED shows boot status for each subsystem (OLED, RTC, SD card, load cells) as it initializes.
2. On successful SD init, a fresh data file is created (`Cal_N.txt`, auto-incremented – see §4) and its name is shown on the OLED, along with the active NAU7802 gain/sample-rate/excitation configuration recorded into the file's header.
3. Apply a known reference weight to the platform and allow the reading to settle.
4. Press **Button C**. The tool reads all three cells' most recently completed background-averaged sample and appends one row to the data file; the OLED shows the running sample count ("**sample N**").
5. Repeat steps 3–4 at as many reference weights as needed for a good calibration curve.
6. Remove the SD card (or copy the file off) and fit the calibration curve offline.

Note

Each capture reads whichever 9-sample background-averaged batch the load cells' own continuous scan most recently completed – it does not trigger a fresh, separately-averaged read. This means the calibration is derived against the exact same read path production weight/thrust recording uses, not a different noise profile.

4 Output File Format

Files are named `Cal_N.txt` (N auto-incremented from 0 each power-up, never overwriting an existing file) in the SD card root. Each file begins with four header lines, then one data row per Button C press:

```

09/09/2026
NAU7802 Gain: 128
NAU7802 Sample Rate (SPS): 320
Excitation Voltage (V): 3.0
Samples Averaged per Reading: 9
Time, Cell1 Raw, Cell2 Raw, Cell3 Raw, Sum Raw, Health
09/09/2026 10:15:03.142, 41230, 40988, 41502, 123720, OK

```

- **Time** – full MM/DD/YYYY HH:MM:SS.mmm timestamp from the RTC.
- **Cell1/2/3 Raw** – raw NAU7802 counts, tare-adjusted, for each of the three cells.
- **Sum Raw** – the three raw values added together (the quantity a combined-weight calibration curve is fit against).
- **Health** – OK if all three cells responded; **DEGRADED** if one or more did not (treat that row's data with caution).

5 Notes

- If the RTC failed to initialize, each row's timestamp is a raw `millis()` count instead of wall-clock time.
- A failed SD write is reported on the OLED ("**write FAILED.**") rather than silently losing the sample – retake it if this happens.
- Status/boot messages are shown on the OLED only; they are *not* written to the data file (the file exists solely to hold calibration rows).

6 Version History

Version	Date	Notes
0.1	September 2026	Initial user guide.